



ANTENEH YEHALEM TEGEGNE

Nationality: Ethiopian **Date of birth:** 7 Apr 1998 **Place of birth:** Bahir Dar, Ethiopia

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ABOUT ME

Machine Learning Engineer with a Master's in AI and Data Science from Bahir Dar University, graduating with distinction. Experienced in Python, PyTorch, and deep learning, with research on sign language recognition and road crash severity prediction. Passionate about applying AI to real-world challenges and seeking opportunities in research, industry, and PhD programs.

WORK EXPERIENCE

Bahir Dar Univeristy – Bahir Dar, Ethiopia

City: Bahir Dar | Country: Ethiopia

LECTURER AND RESEARCH ASSISTANCE (Data Science Department)

[14 Jan 2025 – Current]

Baacumen(Remote) – Seattle, Washington, United States

City: Seattle, Washington | Country: United States

Data Science and Analytics Bootcamp Trainer

[7 Jul 2025 – Current]

Afriwork Company(Remote)

Machine Learning Engineer

[3 Nov 2024 – 4 Jan 2025]

- Collaborating on ML-driven projects focused on real-world applications.
- Contributing to model development, training, and evaluation pipelines.

LINKEDIN, UPWORK – Bahir Dar, Ethiopia

City: Bahir Dar | Country: Ethiopia

FREELANCER

[12 Oct 2023 – Current]

Selam Argiw Kidest Mariyam Acadamy – Bahir Dar, Ethiopia

City: Bahir Dar | Country: Ethiopia

Information Technology Educator

[2 Dec 2022 – 6 Feb 2023]

- maintenance of computers
- relations with suppliers
- coaching a junior Ice Hockey team (10 hours/week)

RESEARCH

[9 May 2023 – 7 May 2024]

End-to-End Continuous Ethiopian Sign Language Recognition

This study tackles continuous Ethiopian Sign Language recognition with an end-to-end model, using a novel dataset of 1,320 sentences from 22 signers. The model integrates 2D and 1D CNNs for spatial-temporal features and BiLSTM for sequence learning, achieving WERs of 8.02% for signer independence and 47.02% for unseen sentences, demonstrating both robustness and areas for improvement.

Link: <https://hal.science/hal-04812102v1>

IMAGE BASED DEEP LEARNING MODEL FOR EARLY DETECTION AND CLASSIFICATION OF COFFEE LEAF DISEASE IN AMHARA REGION

Developed a deep learning and IoT-based system for detecting coffee leaf diseases in Ethiopia. Used 4,300+ real-world images with preprocessing and data augmentation. Fine-tuned CNNs like MobileNetV2 and VGG16, achieving over 97% accuracy. The system provides a scalable solution to support farmers and improve crop productivity.

NeuroScanAI: Deep Learning for Automated Brain Tumor Classification

Developed and evaluated CNN-based models (ResNet50, VGG16, EfficientNetB0, InceptionV3) for classifying brain tumors from MRI scans into four categories. Achieved up to 94% accuracy with ResNet50 and deployed the system as a Flask-based web application for real-time diagnosis, offering a scalable AI tool to support radiology in resource-limited settings.

[30 Dec 2024 – 7 Feb 2025]

Facial Emotion-Driven Music Recommendation

I developed a deep learning-based system that detects facial emotions and recommends music accordingly, using clustering and dimensionality reduction to map emotions to musical attributes via the Spotify API for a personalized experience.

Development of a Model for the Prediction of Road Crash Severity Using Artificial Neural Networks: A Case Study of Bahir Dar City

This study aimed to predict road crash severity using Artificial Neural Networks (ANN) and compare its performance with a multinomial logistic regression model. The research utilized traffic crash data from 2010 to 2022 in Bahir Dar City,

Developing Amharic Sign Language Recognition Model for Amharic Characters Using Deep Learning Approach

This research focused on the recognition of Amharic sign language alphabets using Convolutional Neural Networks (CNN).

EDUCATION AND TRAINING

Msc in Artificial intelligence and Data Science

Bahir Dar Univesity [14 Dec 2021 – 30 Jul 2024]

City: Bahir Dar | Country: Ethiopia | Website: <https://bit.bdu.edu.et/> | Final grade: 3.63 | Thesis: End-to-End Continuous Ethiopian Sign Language Recognition

- Course: Deep Learning, Image processing and Computer Vision, Natural Language Processing (NLP), Research Method and Professional Ethics for AI and DS, Machine Learning, Applied Statistics for AI and DS, Method and Application, Image processing and Computer Vision, Business Intelligence and Data Analytics, Data Privacy and Protection

Bsc Computer Science

Debre Tabor University [13 Mar 2017 – 30 Jul 2021]

City: debre tabor | Country: Ethiopia | Website: <https://dtu.edu.et/> | Final grade: 3.63 | Thesis: Online Hospital Management System (PHP)

- Course: Fundamental of Programming I (C++), Fundamental of Programming II (C++), Linear Algebra, Object-Oriented Programming, Digital Electronics, Fundamentals of Data Base, Advanced Programming, Advanced Database, Internet Programming

Machine Learning Specilization

Deep Learning AI

Website: <https://www.coursera.org/>

Deep Learning Specilization

Deep Learning.AI

Website: <https://www.coursera.org/>

PROJECTS

Breast Cancer Prediction Using Machine Learning Approach

End-to-End Continuous Ethiopia Sign Language Recognition

Named Entity Recognition for Amharic Characters

Part of Speech for Amharic Character using Machine Learning Approach

Online Hospital Management System (PHP)

HONOURS AND AWARDS

[3 Nov 2021 – 4 Jul 2024]

MSc, Scholarship by Bahir Dar University Ethiopia

Received a highly competitive scholarship for the MSc program in Artificial Intelligence and Data Science at Bahir Dar University.

Awardee – Middle East and North Africa Machine Learning (MenaML) Winter School 2025

Selected to attend the competitive MenaML Winter School 2025, focused on cutting-edge machine learning methods in the MENA region.

Awardee – 13th Data Science Africa (DSA) Summer School and Workshop Ibadan, Nigeria

Selected as a participant in the DSA 2025 Summer School and Workshop at Ibadan, Nigeria, covering key topics in AI, data science, and their application to African development.

PROFESSIONAL EXPERIENCE

Organizing Committee Member – 13th Annual Conference on Recent Trends in Scientific Research, Bahir Dar University

Contributed to the planning, coordination, and successful execution of the 13th Annual Conference on Recent Trends in Scientific Research hosted at Bahir Dar University.

Judge, Indaba Ideathon Project Comptition 2024

Served as a judge for the Indaba Ideathon, evaluating projects in AI and data science. Assessed innovative solutions in the field, providing constructive feedback and helping select winners.

Reviewer – African Conference on Computer Vision Summer School (ACVSS 2025)

Served as an Application reviewer for ACVSS 2025

CONFERENCES AND SEMINARS

[6 Jun 2025]

13th Annual Conference on Recent Trends in Scientific Research, Bahir Dar University

Served as an organizing committee member supporting event coordination and logistics. Also participated in the conference by attending paper and poster presentation sessions, engaging with recent developments in scientific research.

INDABA Deep Learning 2025

Attended the 2025 Deep Learning Indaba in Kigali, Rwanda, where I gained valuable insights through keynotes, workshops, presentations, and hands-on practical sessions.

PUBLICATIONS

[2024]

End-to-End Continuous Ethiopia Sign Language Recognition

Authors: Anteneh Yehalem Tegegne

RECOMMENDATIONS

Name: **Birhanu Hailu** | Machine and Deep Learning Teacher

"Anteneh Yehalem exhibited a strong aptitude for AI research, excelling in Machine Learning and Deep Learning courses. His innovative research on Ethiopian Sign Language recognition, utilizing convolutional and recurrent neural networks, showcases his technical skill, intellectual curiosity, and commitment to addressing real-world challenges."

Email: birhanu.hailub@gmail.com

Link: <https://www.linkedin.com/in/birhanu-hailu/>

Name: **Million Meshesha** | Thesis advisor.

"Anteneh Yehalem demonstrated exceptional problem-solving skills, advanced analytical capabilities, and a deep commitment to high-quality research during his MSc studies at Bahir Dar University. His innovative thesis work, academic excellence, and passion for Artificial Intelligence and Data Science make him an outstanding candidate for further research or professional endeavors."

Email: million.meshesha@aaau.edu.et